



THE UNIVERSITY OF JORDAN
KING ABDULLAH II SCHOOL OF INFORMATION TECHNOLOGY
ARTIFICIAL INTELLIGENCE DEPARTMENT

Exploring GAN Variants for Balancing Imbalanced Datasets

Ayham Obeidat

0221882

Project Supervisor:
Dr. yousef sanjalawe

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1 Problem Statement

medical data in real-world often suffer from data imbalance, where one class appears more in the dataset than the other class, for example the dataset used for this project has a substantial imbalance between the two classes where the PNEUMONIA cases appear way more than the NORMAL cases, this imbalance can lead to bias to the majority class and fails to detect the minority class, this issue is crucial for medical diagnostics, as if the detection of the minority class is harder it can lead to delayed or incorrect treatment.

Traditional data augmentation techniques such as rotation, flipping, and scaling are limited in detecting complex visual patterns in medical images, as a result often models trained on imbalance data get high accuracy but poor recall and precision for the minority class.

To address this challenge, this project investigates the use of Generative Adversarial Networks (GANs) to generate realistic synthetic chest X-ray images for the minority class. By augmenting the dataset with GAN-generated samples, the class imbalance problem can be mitigated, potentially improving the generalization and diagnostic performance of classification models.

2 analysis

2.1 Dataset Analysis

the dataset used for this project of chest X-ray images are categorized into two classes NORMAL, and PNEUMONIA, The images are grayscale radiographs

collected from clinical settings and represent pediatric chest X-rays, The dataset is divided into training, testing, and validation subsets to ensure a fair evaluation of model performance.

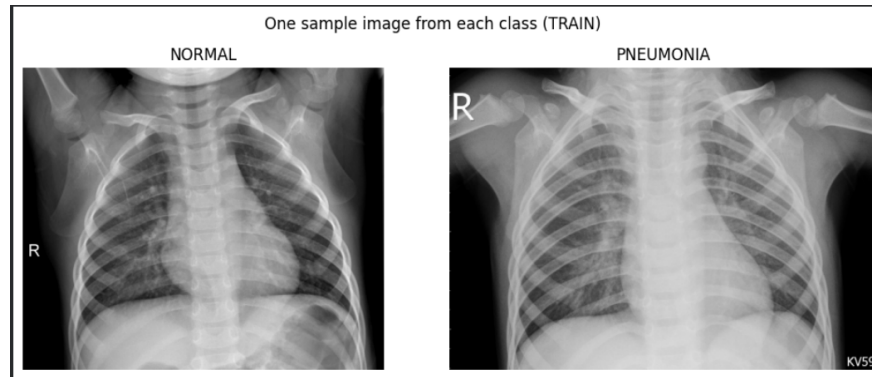


Figure 1: Representative chest X-ray images from the training dataset. The left image shows a Normal chest X-ray with clear lung fields and no visible abnormalities, while the right image illustrates a Pneumonia case characterized by increased pulmonary opacity and structural irregularities. These samples highlight the visual differences between the two classes used in this study.

2.2 Imbalance Analysis

after the analysis there was found to be a huge imbalance between the two class, where the PNEUMONIA has way more cases recorded(3875 cases) that the NORMAL cases(1341 cases).

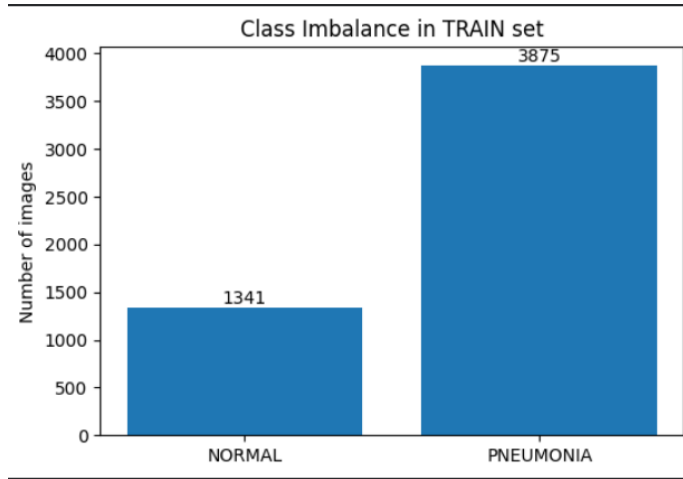


Figure 2: Class distribution in the training dataset. The figure illustrates a pronounced class imbalance between the Normal and Pneumonia classes, where Pneumonia cases (3,875 images) significantly outnumber Normal cases (1,341 images). This imbalance can bias classification models toward the majority class and motivates the use of data augmentation and GAN-based synthetic sample generation.

3 GAN Architectures and Training

3.1 Overview Of GAN Framework

Generative Adversarial Networks (GANs) consist of two competing neural networks: a Generator (G) and a Discriminator (D). The generator learns to produce synthetic images that resemble real samples, while the discriminator attempts to distinguish between real and generated images. Through adversarial training, both networks improve iteratively, enabling the generator to produce increasingly realistic images.

in this project we used GANs to generate images of chest x-rays for the minority class to address the imbalance in the data.

3.2 Vanilla GAN (DCGAN) Architecture

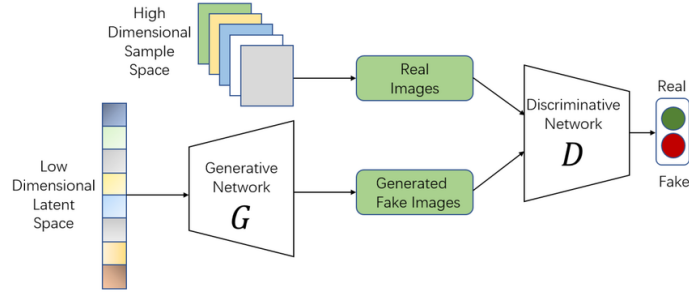


Figure 3: The architecture of vanilla GANs

3.2.1 Generator Architecture(DCGAN)

The generator in the DCGAN model takes a random noise vector sampled from a standard normal distribution as input. This noise vector is progressively up-sampled using a series of transposed convolutional layers to generate a synthetic chest X-ray image.

3.2.2 Discriminator Architecture

The discriminator receives an image and predicts whether it is real or synthetic. It consists of convolutional layers that progressively downsample the input image and extract hierarchical features.

3.3 Conditional GAN (CGAN) Architecture

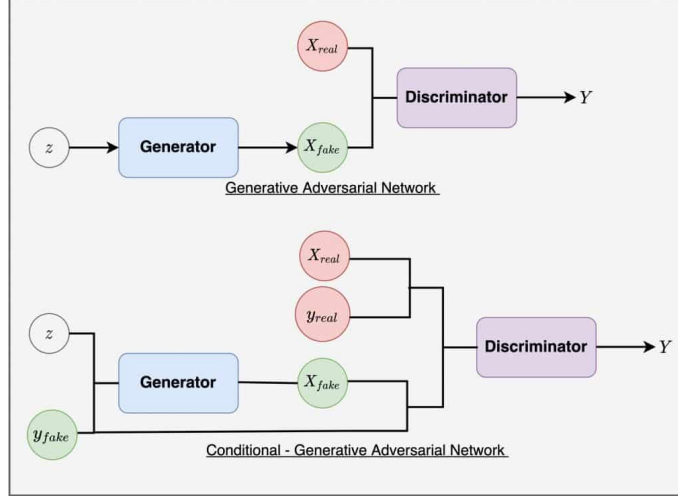


Figure 4: The architecture Conditional GAN

3.3.1 Conditional Generator

The Conditional GAN extends the vanilla GAN by incorporating class information into the generation process. In this model, the generator receives both a noise vector and a class label, which is embedded and concatenated with the noise vector before being passed through the network.

3.3.2 Conditional Discriminator

The conditional discriminator receives both the image and its corresponding class label. The label is transformed into a spatial map and concatenated with the image, allowing the discriminator to verify whether the image-label pair is consistent.

3.4 Training Strategy

3.4.1 Dataset Usage

Both GAN models were trained exclusively on the minority class to generate synthetic samples representative of underrepresented data. This ensures that generated images specifically augment the minority class without altering the majority class distribution.

3.4.2 Loss Function

Binary Cross-Entropy loss was used for both the generator and discriminator. The discriminator is optimized to correctly classify real and synthetic images, while the generator is optimized to maximize the discriminator’s likelihood of miss-classifying generated images as real.

3.4.3 Optimization

The Adam optimizer was used for both networks with the following parameters, learning rate of $2e-4$, and β_1 set to 0.5 and β_2 set to 0.999, These settings are commonly adopted in GAN training and provide stable convergence.

3.4.4 Training Configuration

the image size was set to 64 by 64, the batch size set to 64, Number of training epochs was 30, the input was normalized to pixel values scaled to $[-1,1]$, and Fixed noise vectors were used during training to monitor the progression of generated image quality across epochs.

4 Results and Comparisons

The performance of the classifier was evaluated using multiple metrics which are, Accuracy, Precision, Recall, F1-Score, Area Under the ROC Curve (AUC-ROC), and Confusion Matrix, These metrics are particularly important in imbalanced medical datasets, where accuracy alone may not reflect true model performance.

4.1 Results

Table 1: Classification performance comparison across different dataset configurations

Scenario	Accuracy	Precision	Recall	F1-score	AUC
Imbalanced Dataset	0.74	0.72	0.98	0.83	0.87
Balanced (DCGAN)	0.75	0.72	0.99	0.83	0.86
Balanced (CGAN)	0.80	0.77	0.98	0.86	0.87

The results in Table 1 reveal important differences in classifier behavior across the three training scenarios. The model trained on the original imbalanced dataset achieves a relatively high recall of 0.98, indicating strong sensitivity toward Pneumonia cases; however, this is accompanied by lower precision (0.72), suggesting a higher rate of false positives. When the dataset is balanced using DCGAN-generated images, recall remains high (0.98), but slight decreases in accuracy and F1-score are observed, indicating that while minority detection is preserved, overall classification performance does not significantly improve. In

contrast, balancing the dataset using Conditional GAN leads to the best overall performance, achieving the highest accuracy (0.80), precision (0.77), and F1-score (0.86), while maintaining a high recall (0.97) and AUC of 0.88. These results suggest that CGAN-based augmentation produces higher-quality synthetic samples that enhance class discrimination and improve model generalization without sacrificing sensitivity to Pneumonia cases.

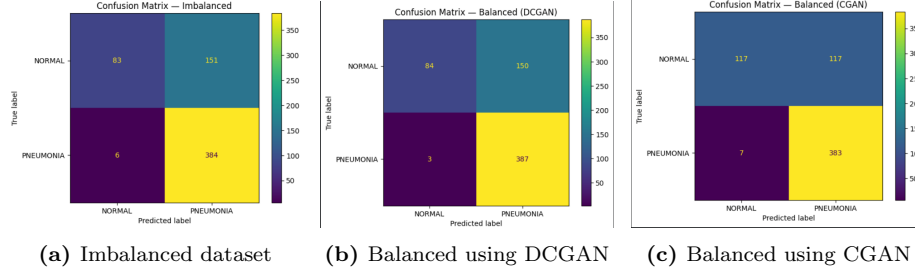


Figure 5: Confusion matrices for the three training scenarios. The matrices compare classification performance on the original imbalanced dataset and datasets balanced using DCGAN and Conditional GAN. GAN-based balancing, particularly with CGAN, improves the correct classification of the Normal class while maintaining high detection rates for Pneumonia cases.

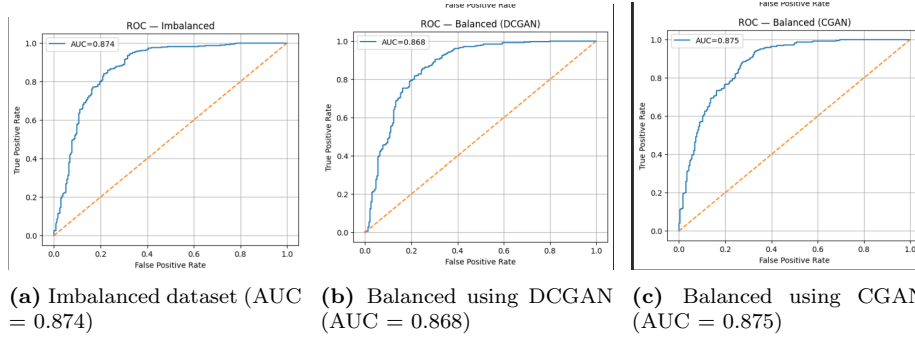


Figure 6: ROC curves for the three training scenarios. The ROC curves compare the trade-off between true positive rate and false positive rate for classifiers trained on the original imbalanced dataset and datasets balanced using DCGAN and Conditional GAN. GAN-based balancing results in improved AUC values, indicating enhanced discriminative capability compared to the imbalanced scenario.

4.2 Comparisons

Overall, balancing the dataset using GAN-generated images resulted in improved classification performance, particularly in terms of recall and F1-score for the minority class. Among the two GAN approaches, the Conditional GAN

demonstrated slightly superior performance, suggesting that conditional information can improve the quality and usefulness of generated samples.

5 Observations and Conclusions

This project investigated the impact of GAN-based data augmentation on addressing class imbalance in chest X-ray image classification. The original dataset exhibited a significant imbalance between Normal and Pneumonia classes, which biased the classifier toward the majority class and limited its overall generalization. To mitigate this issue, two GAN-based approaches—Vanilla GAN (DCGAN) and Conditional GAN (CGAN)—were employed to generate synthetic images for the minority class.

Experimental results demonstrate that GAN-based augmentation improves classification performance compared to training on the original imbalanced dataset. While the imbalanced dataset achieved high recall for Pneumonia cases, it suffered from lower precision and reduced recognition of Normal images. Balancing the dataset using DCGAN preserved high sensitivity but provided limited improvement in overall performance. In contrast, the CGAN-balanced dataset achieved the best overall results, yielding higher accuracy, precision, F1-score, and competitive AUC-ROC values. Confusion matrix and ROC curve analyses further confirmed that CGAN-based augmentation enhanced class discrimination without significantly increasing false negatives.

Overall, the findings indicate that GAN-based synthetic data generation is an effective strategy for mitigating class imbalance in medical imaging tasks. In particular, Conditional GANs demonstrate superior capability in producing informative synthetic samples that improve classifier robustness and generalization. These results highlight the potential of GAN-based augmentation as a practical tool for improving performance in data-limited and imbalanced medical datasets.